

Terrain-Aware, Latency-Robust Control for Off-Road Autonomy and Teleoperation

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Abstract

We present a real-time, latency-aware human-in-the-loop framework for safe shared and autonomous control of vehicles on deformable Soil Contact Model (SCM) [4] terrain. A differentiable neural surrogate of the SCM tire forces is embedded inside an acados NMPC for terrain-aware trajectory tracking, while a learned online estimator regresses the dominant soil parameters from onboard IMU and wheel-speed signals so that both the NMPC and a latency-aware DOB-CBF-QP safety filter track the terrain and compensate time-varying teleoperator delay in real time. We simulate an HMMWV on SCM terrain using Chrono::Vehicle (Project Chrono [7]); Chrono::Sensor supplies driver POV and a Logitech G29 steering/pedals. Simulator, NMPC, terrain estimator, and CBF-QP exchange data over low-latency ZMQ; ML-based 5G models [2] inject delay only on the camera and human-control paths, not between simulator and controller.

Contributions. Off-road teleoperation must contend with deformable-soil effects [1] that rigid-terrain tire models cannot capture without per-deployment recalibration [3], while time-varying network latency [2] requires latency-aware shared-control arbitration [8]. Building on Project Chrono [7] and the human-in-the-loop framework Chrono::HIL [9], we contribute: (i) a differentiable neural SCM tire surrogate that augments the static MLP of [3] with rate-augmented and ResNet variants and exposes a peak longitudinal traction budget F_x^{trac} to the NMPC as $|Ma_x| \leq F_x^{\text{trac}}$ (M vehicle mass, a_x commanded longitudinal acceleration), paired with an asymmetric throttle DOB on the actuation map that closes the residual soft-soil speed gap left by open-loop $\dot{u} = a_x$ with u longitudinal speed; (ii) a learned online terrain estimator that replaces the UKF of [3] and jointly regresses Bekker n and Mohr–Coulomb ϕ from onboard IMU and wheel-speed signals, conditioning both the surrogate and the traction-aware NMPC constraints online; (iii) a two-layer obstacle-avoidance stack that pairs in-horizon NMPC softplus barriers with a downstream latency-aware DOB-CBF-QP safety filter [8] whose friction cone is tightened by the same ϕ estimate; and (iv) an open-source parallelized data-collection and benchmarking suite that scales training and closed-loop evaluation across terrains and reference trajectories.

Technical content

Human-in-the-loop stack. Chrono::Vehicle [5] runs an HMMWV on an SCM patch within Chrono::HIL [9]. The human views Chrono::Sensor driver POV and steers a Logitech G29 via pygame with SDL2 haptics. IMU measurements ride on the ZMQ vehicle-state stream to NMPC and the terrain estimator. A second process runs NMPC and CBF-QP on low-latency ZMQ to the sim; [2] delays hit the simulated camera stream and operator commands (Fig. 1); one workstation closes the loop in real time.

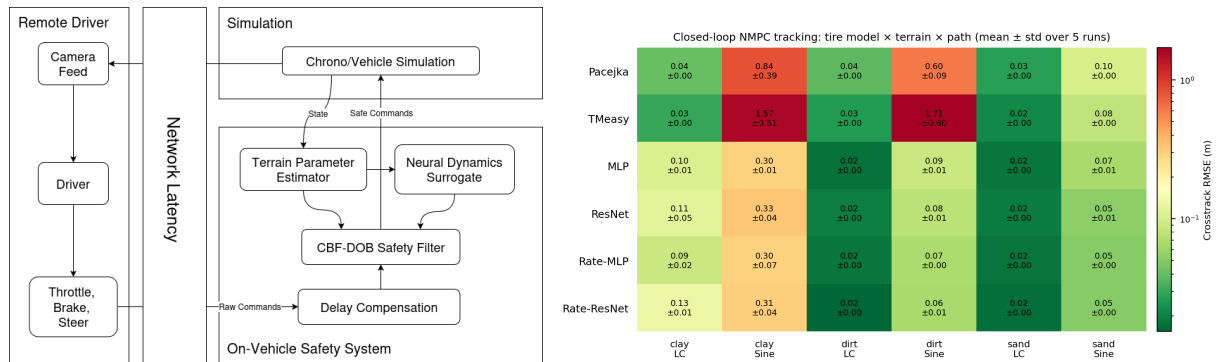


Figure 1: **Left:** shared-control architecture: SCM surrogate and learned terrain estimator inform the NMPC and the DOB-CBF-QP filter that screens delayed operator inputs. **Right:** NMPC tracking at 5 m/s (LC and sinusoid, 2 m amp, 30 m wavelength); RMS crosstrack error (m), mean±std, 5 runs/cell.

Differentiable SCM surrogate. A small neural network maps (κ, α, F_z, u) , steering rate, and soil parameters $\theta_{\text{soil}} = (K_\phi, K_c, n, c, \phi, k)$ (Bekker–Wong–style rig inputs [6]) to longitudinal and lateral tire forces. Where [3] employ a static MLP, we add rate-augmented and ResNet variants (Fig. 1, right); all train on parallelised Project Chrono SCM tire-rig sweeps and export as CasADi expressions so acados auto-differentiates through the tire forces every solve step. The longitudinal traction cap F_x^{trac} queries the surrogate at reference slip κ_{ref} with $\alpha = 0$, using axle loads F_z , speed u , and θ_{soil} from terrain presets or the online estimator. Cornering forces F_y use the same θ_{soil} but the live operating point (κ, α, F_z, u) at each NMPC stage.

Online terrain estimation. Where Dallas et al. [3] drive an Unscented Kalman filter through their NN tire model, we train a window MLP that regresses Bekker n and Mohr–Coulomb ϕ from a few seconds of onboard IMU and wheel-speed history, using dimensionless coupling channels that make the descriptor approximately speed-invariant. Trained on a mixed open- and closed-loop corpus, the estimator converges within seconds on canonical clay, dirt and sand (Fig. 2), generalises to randomly sampled unseen soils, and re-conditions the surrogate online so that closed-loop tracking improves by an order of magnitude over Pacejka and TMeasy (Fig. 1, right). Rich-excitation LHS data support joint (n, ϕ) regression (held-out RMSE ≈ 0.07 in n , $\approx 3.3^\circ$ in ϕ); ensemble disagreement on ϕ gates the safety filter’s friction cone.

NMPC barriers and delay-aware safety filter. Obstacle handling uses two complementary layers. (i) The NMPC adds a smooth softplus barrier (stage and terminal) over live- ϕ traction limits, proactively re-planning inside its horizon. (ii) A downstream DOB-CBF-QP safety filter [8] screens autonomous and operator commands via a heading-aligned ellipsoidal barrier elongated ahead of the vehicle: a filter-side first-order velocity DOB captures unmodelled SCM drag, the NN surrogate sets the traction-limited top speed, and the barrier plus forward-prediction collision check are inflated by an EMA teleoperation round-trip delay drawn from the same 5G model [2].

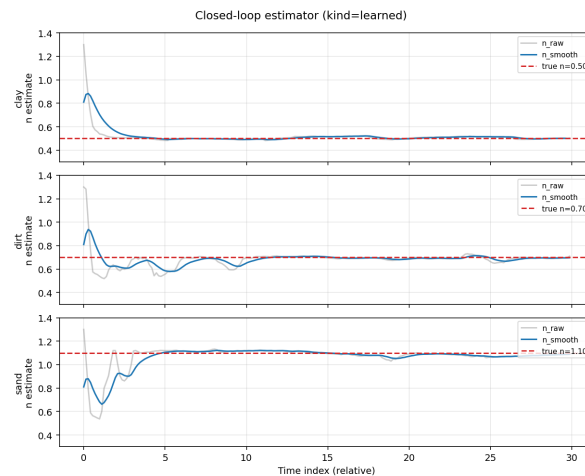


Figure 2: Convergence of $\hat{n}$ on canonical clay/dirt/sand under closed-loop NMPC at 5 m/s.

Open-source data and benchmarks. Parallelized SCM sweeps for training and a benchmark suite across terrains and reference trajectories; per-process ZMQ enables running the Fig. 1 6×6 grid (180 NMPC trials) in under 30 min on one workstation.

References

- [1] Bekker, M.G.: Introduction to Terrain-Vehicle Systems. University of Michigan Press, Ann Arbor, 1969.
- [2] Choi, Y.-H. et al.: ML-based 5G traffic generation for practical simulations using open datasets. IEEE Commun. Mag., Vol. 61, pp. 130–136, 2023. doi: <https://doi.org/10.1109/MCOM.001.2200679>.
- [3] Dallas, J. et al.: Terrain adaptive trajectory planning and tracking on deformable terrains. IEEE Trans. Veh. Technol., Vol. 70, No. 11, pp. 11255–11268, 2021. doi: <https://doi.org/10.1109/TVT.2021.3114088>.
- [4] Krenn, R.; Hirzinger, G.: SCM – a soil contact model for multi-body system simulations. ISTVS, Bremen, 2009.
- [5] Serban, R., Taylor, M., Negrut, D., Tasora, A.: Chrono::Vehicle: template-based ground vehicle modelling and simulation. Int. J. Veh. Perf., Vol. 5, No. 1, pp. 18–39, 2019.
- [6] Shibly, H., Iagnemma, K., Dubowsky, S.: An equivalent soil mechanics formulation for rigid wheels in deformable terrain. J. Terramech., Vol. 42, No. 1, pp. 1–13, 2005. doi: <https://doi.org/10.1016/j.jterra.2004.05.002>.
- [7] Tasora, A. et al.: Chrono: an open source multi-physics dynamics engine. HPCSE, pp. 19–49, Springer, 2016. doi: https://doi.org/10.1007/978-3-319-40361-8_2.
- [8] Zhang, H. et al.: Shared control of teleoperated vehicles with delay-compensated safety filtering. IEEE CCTA, pp. 192–199, 2025.
- [9] Zhou, Z. et al.: A Chrono-based framework for large-scale traffic simulation with human-in-the-loop. 2023. doi: <https://doi.org/10.13140/RG.2.2.23133.59361>.